Rongxuan Zhou

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Research Interests

My research focuses on robot learning for contact-rich manipulation, in particular on world models that know the limits of their own perception and action. It follows three threads: latent world models that plan from vision and force, and the failure modes that arise when a sensing modality or an action direction is absent from the model; training-free safety and diagnostic methods that act on frozen checkpoints at sampling time; and the deployment of these models on real manipulators and humanoids.

Education

- **Northeastern University** *Sep 2024 – May 2026*
  M.S. in Robotics, Electrical and Computer Engineering — Advisor: Prof. Yang Ye Boston, MA
  - Thesis: *TACS: Training-Free Safety Guidance for Action-Space Diffusion Policies via Tweedie-Adaptive Constrained Sampling*
- **Northeastern University** *Sep 2020 – Jun 2024*
  B.E. in Mechanical Engineering and Automation Shenyang, China
  - T-DT Lab mechanical design lead; First Prize, 2021 RoboMaster University Championship

Research Experience

- **Northeastern University** *Feb 2025 – Present*
  Graduate Research Assistant — Advisor: Prof. Yang Ye Boston, MA
  Research on world models and safe policy learning for contact-rich manipulation. I lead two first-author lines (force fusion, action-space blind spots), co-develop the lab's other world-model projects, and own the lab's real-robot stack: controller, teleoperation, digital twin and data collection for the Franka FR3 and the humanoid experiments.
  - **ForceLeWM: A Modality-Robust JEPA World Model with Gated Force Fusion**
    - Built a JEPA world model that fuses force/torque through a learned per-dimension gate: on simulated PushT-FT with force absent at test time, naive fusion cut planning success from 41% to 15%; gated fusion with alignment restored it to 35%.
    - Deployed one frozen sim-trained checkpoint per task on real Franka assembly, seating the part in 27 of 30 gear-meshing and 29 of 30 peg-insertion trials, with force from a proprioceptive wrench observer instead of a dedicated F/T sensor.
  - **Subspace-Restricted Planning under Action Compression in Latent World Models**
    - Showed that action compression leaves latent-world-model planners partly blind: on a released Cube checkpoint 15 of 25 action dimensions never reach the predictor, so CEM executes random motion along them that no model-side check detects.
    - Proposed training-free pruned sampling from one SVD of the frozen action encoder, provably unable to worsen the best plan, plus an active-subspace cut on the cost-gradient spectrum that raised success 10.7 points on Cube and 12.3 on Cube-double.
  - **TACS: Training-Free Safety Guidance for Action-Space Diffusion Policies**
    - Added workspace safety to pretrained diffusion policies without retraining, by injecting barrier-function guidance into the denoising loop: 20× fewer constraint violations on robosuite Lift (6.6% → 0.32%, $p = 0.015$, 5 seeds).
    - Built the Franka FR3 deployment path: a 1 kHz Cartesian impedance controller (C++, ROS 2, libfranka) interpolating 10 Hz policy action chunks, with barrier guidance applied to Tweedie-predicted end-effector positions inside the inference loop.
  - **PRISM: Prior-Guided Imagination Sampling in World Models**
    - Built the teleoperation and data stack on a Franka FR3, streaming Meta Quest 3 (WebXR) commands through a 1 kHz Cartesian impedance controller, and open-sourced a 500-demonstration PushT dataset collected with it.
    - Trained the JEPA latent world model (ViT encoder + action-conditioned autoregressive Transformer) on that data and deployed it closed-loop on the robot with real-time MPPI sampling-based planning in the latent space.
  - **TOIF: Terrain-Aware Hierarchical Control with Proprioceptive Feedback for Humanoid Navigation**
    - Co-developed the feedback path of a three-tier humanoid navigation stack: proprioceptive anomaly detectors feed leaky integrators whose output becomes structured language, so a high-level planner replans over a semantic scene graph while a VLA model issues velocity commands.
    - Ran the Unitree G1 hardware validation across uneven, stepped and slippery terrains: transferred the Isaac Lab-trained anomaly detectors to the robot (AUC 0.87 and 0.94 on rough and stepped terrain) and closed the loop to replanning.
  - **EquiJEPA: Learning Action Representations for Effect Prediction and Action Generation**
    - Co-developed the state-blind effect head of EquiJEPA, which makes the same action yield one shared latent transition across states: transition concentration rose from 0.747 to 0.910 on Push-T and from 0.570 to 0.872 on Cube.
    - Evaluated the learned action prototypes by physical execution, with cross-state reuse and unseen action composition: agent error at new Push-T states fell from 4.02 to 2.67 px and two-dot composition success rose from 11.5% to 40.1%.
  - **What Reveals Below: Fold-Contact Affordances for Cloth Unfolding**

- Co-developed fold-contact affordances for dual-arm cloth unfolding: a transparent table gives a top mask and a bottom contact mask, converted into fold residual, contact overlap and frontier maps that mark where hidden folds meet grounded support.
- Built the physical transparent-table dual-arm rig with two RealSense D435 cameras in top and bottom views, and ran the real-robot unfolding trials with the garment state segmented by SAM 3 from synchronized observations.

- **Shanghai Jiao Tong University, IWIN-FINS Lab** *Jun 2022 – Aug 2024*
  Research Assistant — Advisor: Prof. Jianping He Shanghai, China
  - **FINES: Full-Stack Mobile Manipulation Platform — Hardware-Software Co-Design**
    - Architected the control stack from bare metal up (Master MCU, Intel NUC, hardware watchdog) with a C++ embedded framework (FineMote) managing CAN/RS485/I2C at a 1 ms cycle, supporting 3 downstream research projects.
    - Designed a custom 6-DOF manipulator: actuator and transmission sizing, rigid-body dynamics, Pieper-condition closed-form IK, and MoveIt integration, closing the loop from CAD to real-time joint-space control on the embedded stack.
    - Built the perception pipeline end-to-end: multi-camera calibration with bird's-eye-view homography, Point-LIO LiDAR-inertial odometry fused with UWB indoor positioning, and YOLO + ICP obstacle detection for autonomous navigation in clutter.
    - Extended the platform to multi-agent coordination: synchronized the manipulator with the AGV over ROS 2 for mobile manipulation, and built an air-ground transport system with MPC-based quadrotor control validated hardware-in-the-loop.
- **Shanghai University, Key Lab of Intelligent Manufacturing and Robotics** *Jun 2022 – Apr 2023*
  Research Assistant — Advisor: Prof. Guiqin Li Shanghai, China
  - Developed ANSYS/ABAQUS finite-element models of swage-autofrettage high-pressure cylinders, capturing mandrel interference and elasto-plastic loading with the Bauschinger effect to predict the residual-stress field.
  - Swept mandrel geometry in a parametric ABAQUS contact model to optimize the protective compressive residual-stress distribution and fatigue life of the cylinder, with mesh-convergence studies and FEA results within 5% of experimental data.

Industry Experience

- **XtalPi** *May 2026 – Present*
  Robotics Software Engineer, Controls and Robot Learning — Full-time Cambridge, MA
  Built the control, simulation and learning stack of an autonomous chemistry lab delivered to Eli Lilly's US site and running 24 station modules through up to 144 unattended reactions per order: experiment scheduling, arm control, vision-guided dosing, an LLM operations agent and device middleware.
  - Built the vision and simulation-learning loop for milligram powder dosing: two-camera MobileNet-UNet segmentation calibrates each spatula tip to a 6-DoF pose in the live dosing flow, and a Blender scoop simulation trains a Rainbow DQN policy as a learned alternative to the rule-based controller.
  - Implemented motion control for five CANopen arms: pose statemaps with three-point affine teaching, PRM planning over safe zones exported from Blender layouts and MJCF collision models, and a browser digital twin (prototype) that rehearses the 8 ms joint trajectory the real arm then replays.
  - Built the experiment scheduler around a discrete-event model of the station: one simulation generates each order's process graph, scores ALNS plate layouts, returns per-plate ETAs and lets a follow-on order start before the last one ends.
  - Developed the LLM operations agent for the station: 52 tool capabilities that validate experiment designs, commit orders to the scheduler and triage failures from logs, with every state-changing action behind a token gate; validated by 109 behavior evals and 636 unit tests.
  - Architected the device middleware: 17 driver types over eCAL and Cap'n Proto for CAN, CANopen, Modbus and serial hardware, a module framework exposing 311 async ops, and a state machine that turns raw bus data into module states and E-stop events.
- **Dinnar Automation** *Apr 2025 – Dec 2025*
  Robotics & Machine Vision Engineer — Contract, Hybrid San Jose, CA
  Full-stack engineering contributor to the CSE Round Lighting Check & Final-Inspection (FI) AOI machine, a vision-guided robotic inspection system for Texas Instruments Chip Scale Element (CSE) products, in production at 85K+ units/day.
  - Implemented the vision-guided inspection pipeline: an Epson SCARA robot (dual-nozzle, 4 CSE per cycle, Poka-Yoke orientation check), servo-driven part reorientation, and 4-angle hardware-triggered CCD inspection, covering 19 defect categories at 100% detection on acceptance samples.
  - Wrote the classical-CV defect detectors in C++: NCC template matching with edge and blob checks for cracks, HSV thresholds for contamination, OCR with a sharpness measure for markings, and a dark-field test for light leakage.
  - Reached a 3.8 s takt per four-part batch by imaging the bottom face during transfer under strobe instead of at a stop, and added an NG reconfirmation camera that re-images flagged parts off the critical path to cut false rejects.
- **CloudMinds Robotics** *Dec 2023 – Apr 2024*
  Robotics Control Engineer — Internship, On-site Shanghai, China
  Contributed real-time control infrastructure for multi-robot assembly cells used in harmonic-reducer production, plus a data-driven UR5e impedance controller, bridging factory-grade automation with manipulation research.
  - Rebuilt the manipulator's real-time communication stack on EtherCAT with standards-compliant IEC 61131-3 Structured Text, cutting the joint control cycle by 40% and delivering sub-millimeter trajectory repeatability measured per ISO 9283.
  - Replaced the analytical Hogan impedance model with a data-driven controller on UR5e for contact-rich assembly, achieving 42% vibration reduction in a 72-hour continuous aging test against the analytical baseline on the same assembly task.

- Built an OPC UA bridge between ROS 2 and the plant's Siemens S7-1500 PLCs, so research controllers could run on the production PLC network at $\leq \pm 2$ ms jitter with no custom protocol on the PLC side and no changes to the plant stack.

- **NIO Automobile** *Jul 2023 – Oct 2023*
  Autonomous Driving Research Intern — Internship, On-site Shanghai, China
  Research internship on the autonomous-driving team: generative data augmentation for ECU testing and sim-to-real transfer for vehicle lateral control.
  - Generative augmentation for safety-critical edge cases: built a GAN-based CAN-bus signal synthesizer producing 5× rare-event training data, expanding ECU long-tail test coverage by 35% and surfacing failure modes classical fuzzing missed.
  - Domain-adversarial sim-to-real policy transfer (DANN): implemented gradient reversal layers to align CARLA and NIO ET5 feature distributions; closed-loop field-tested on real roads, reducing lateral-control error by 37%.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION,†=FIRST AUTHOR

[S.6] **R. Zhou**†, Y. Wang, X. Hu, and Y. Ye (2027). **ForceLeWM: A Modality-Robust JEPA World Model with Gated Force Fusion**. In submission, *IEEE International Conference on Robotics and Automation* (*ICRA*).

[S.5] **R. Zhou**†, Y. Wang, X. Hu, S. Cao, and Y. Ye (2027). **Subspace-Restricted Planning under Action Compression in Latent World Models**. In submission, *International Conference on Learning Representations* (*ICLR*).

[S.4] Y. Wang, J. Xia, B. Sun, **R. Zhou**, X. Hu, Y. Shi, J. Du, and Y. Ye (2027). **PRISM: PRior-guided Imagination Sampling in World Models**. arXiv preprint 2026; in submission, *IEEE International Conference on Robotics and Automation* (*ICRA*).

[S.3] Y. Wang, H. Zhang, Q. Yin, **R. Zhou**, X. Hu, Y. Shi, and Y. Ye (2027). **EquiJEPA: Learning Action Representations for Effect Prediction and Action Generation**. In submission, *International Conference on Learning Representations* (*ICLR*).

[S.2] **R. Zhou**† and Y. Ye (2026). **TACS: Training-Free Safety Guidance for Action-Space Diffusion Policies via Tweedie-Adaptive Constrained Sampling**. In submission, *IEEE Robotics and Automation Letters* (*RA-L*). Master's thesis.

[S.1] X. Hu, L. Sun, Y. F. Yeung, Y. Wang, **R. Zhou**, and Y. Ye (2026). **What Reveals Below: Fold-Contact Affordances for Cloth Unfolding**. In submission, *Conference on Robot Learning* (*CoRL*).

[C.2] Y. Wang, **R. Zhou**, C. Wang, X. Li, Z. Teng, X. Hu, A. Ramezani, J. Du, and Y. Ye (2026). **Thinking on Its Feet: Terrain-Aware Hierarchical Control with Proprioceptive Feedback for Humanoid Navigation**. In *IEEE-RAS International Conference on Humanoid Robots* (*Humanoids*).

[C.1] W. Xi, J. Guo, C. Wang, S. Wu, and J. He (2025). **A Robust Distributed Odometry for Mobile Robots with Steerable Wheels**. In *IEEE/RSJ International Conference on Intelligent Robots and Systems* (*IROS*). Contribution: robot structural and simulation developer.

[J.1] L. Sun, **R. Zhou**, G. Li, J. Li, and P. Mitrouchev (2023). **Simulation Research on Residual Stress of Swage Autofrettage-processed High-Pressure Cylinder**. *Journal of Physics: Conference Series*, Vol. 2587, 012088 (ICAMMT-2023).

OPEN-SOURCE SOFTWARE

- **frlink — real-time control and teleoperation stack for Franka FR3** *2026*
  C++ / Python, libfranka @Rongxuan-Zhou/frlink
  - 1 kHz torque-level Cartesian impedance controller on a real-time host, with a client side for Meta Quest 3 (WebXR) teleoperation and policy action chunks; 0.10 mm drift and zero missed cycles in a 5-minute hold.

- **MuJoCo MCP Server** *2025*
  Python, MuJoCo, Model Context Protocol @Rongxuan-Zhou/mujoco-mcp-server
  - A 65-tool Model Context Protocol server that lets LLM agents drive MuJoCo simulation through natural language: scene and robot loading for 50+ models from MuJoCo Menagerie, inverse kinematics, trajectory optimization (iLQR, MPPI), domain randomization and RL environment integration.

SKILLS

- **Robot Learning:** Diffusion Policy, World Models (JEPA), Imitation Learning, Reinforcement Learning (Rainbow DQN), VLA, sampling-based planning (MPPI, CEM), LLM agents (tool use)
- **Real-Robot Systems:** Franka FR3, UR5e, Unitree G1, teleoperation (WebXR, Meta Quest 3), sim-to-real, force/torque sensing, ROS 2, CANopen, EtherCAT, eCAL, MoveIt, FANUC, ABB
- **Control:** 1 kHz torque control, Cartesian impedance control, kinematics (FK / IK), rigid-body dynamics, trajectory tracking, PRM motion planning, PLC programming (IEC 61131-3)
- **Perception & Vision:** Multi-camera and hand-eye calibration, LiDAR-inertial odometry (Point-LIO), YOLO, U-Net segmentation, SAM, ICP, OpenCV, AOI system design
- **Simulation:** MuJoCo (MjSpec), Isaac Sim / Isaac Lab (PhysX), Genesis, pymunk, Blender, MJCF, digital twins, domain randomization, discrete-event simulation (SimPy)
- **Hardware & Embedded:** EtherCAT, CANopen drives (CiA 402), STM32, Jetson, SolidWorks, ANSYS, ABAQUS, CODESYS, Siemens PLC, OPC UA, CAN / RS485 / I2C
- **Programming & Infra:** Python, PyTorch, CUDA, C/C++, MATLAB, Linux, Git, Docker, CI/CD, Hugging Face datasets